Deepfake Text Detection Systems for Arabic and English Using BERT and ELECTRA-Based Models

CS4083 - Natural language processing & text mining

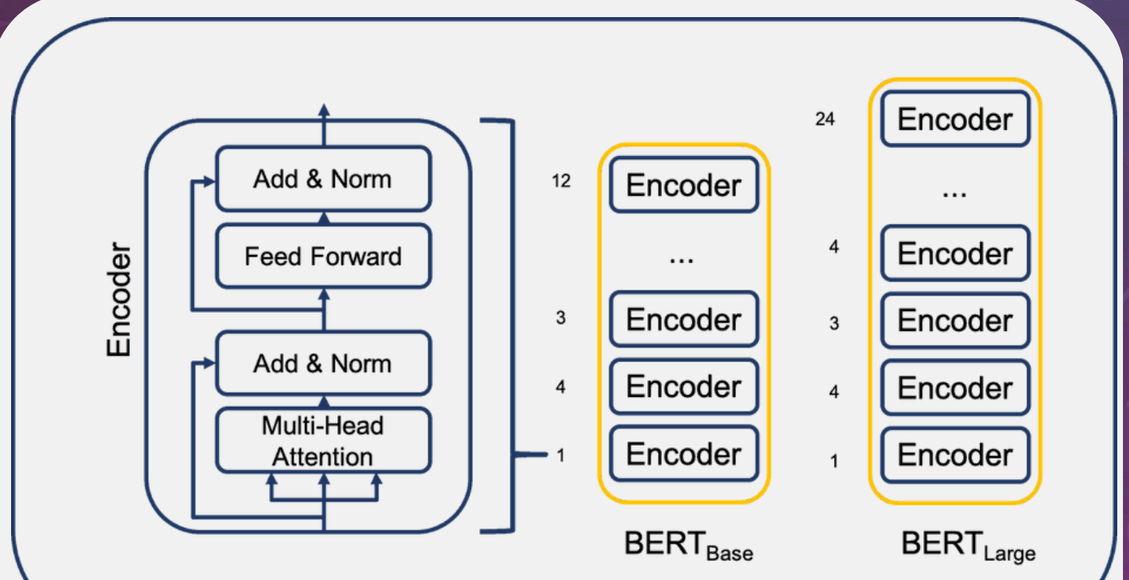


The 23 International Learning and Technology Conference

Twin Minds: Agentic AI and the New Era of Autonomy

توأمة العقول: الذكاء الاصطناعي الوكيل لأتمتة الأعمال

MODELS USED



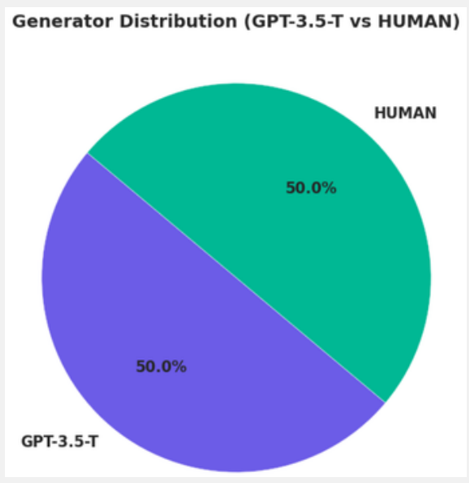
- MODELS FOR DETECTING ARABIC: ARAELECTRA - ARABERT LARGE - ARABERT BASE
- MODELS FOR DETECTING ENGLISH: DISITBERT - ENGBERT - DEBERTA-V3-BASE

INTRODUCTION

AI-generated text, produced by tools like ChatGPT, Gemini, and Claude, is now widespread in many areas, making it increasingly difficult to distinguish from human-written content.

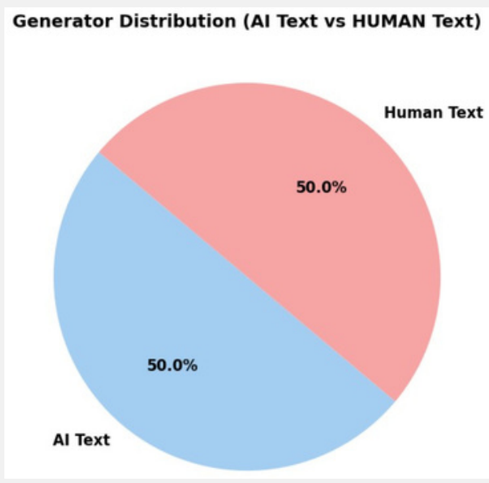
DATASETS

Arabic Dataset



ALHD

English Dataset



BALANCED_AI_HUMAN_PROMPTS

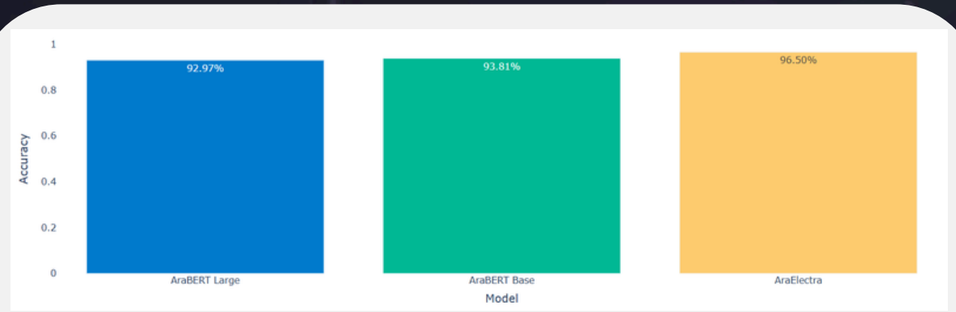
Preprocessing (Arabic)

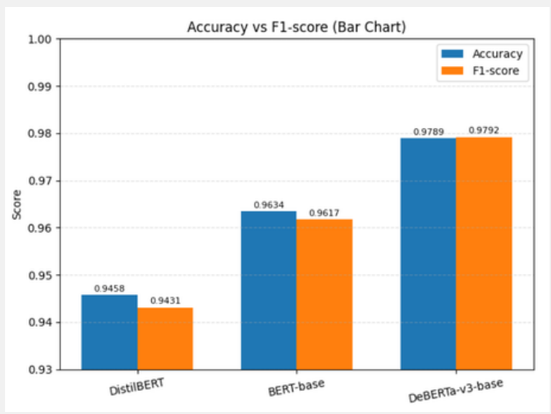
Step	Description
Load Dataset & Basic Exploration	Already clean and no missing values
Spliting the Dataset	72% training - 20% testing - 8% validation
Tokenization of Text	• Text → input IDs and attention masks • Truncation: Long text is shortened to 128 tokens • Padding: All sequences are aligned to the same length • Normalization handled automatically by tokenizer
Creating a PyTorch Dataset	This prepares the data for batching during training.
Preparing Text for the Model	Converting each input into tensors

Preprocessing (English)



ARABIC/ ENGLISH MODELS RESULT





INPUT:

تعتزم الشركة إطلاق نظام ذكاء اصطناعي متقدم قادر على تحليل النصوص العربية بدقة عالية، وقد تم تدريب النموذج على ملايين الجمل، لكن تقارير أخرى تدعي أن النظام يتسبب في أخطاء أخلاقية غير مقصودة

OUTPUT:

Model	label	fake_probability
AraBERT-base	FAKE	0.9955
AraBERT-large	FAKE	0.9814
AraElectra	FAKE	0.9996

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